

TOPIC: CS2 LAN INFRASTRUCTURE

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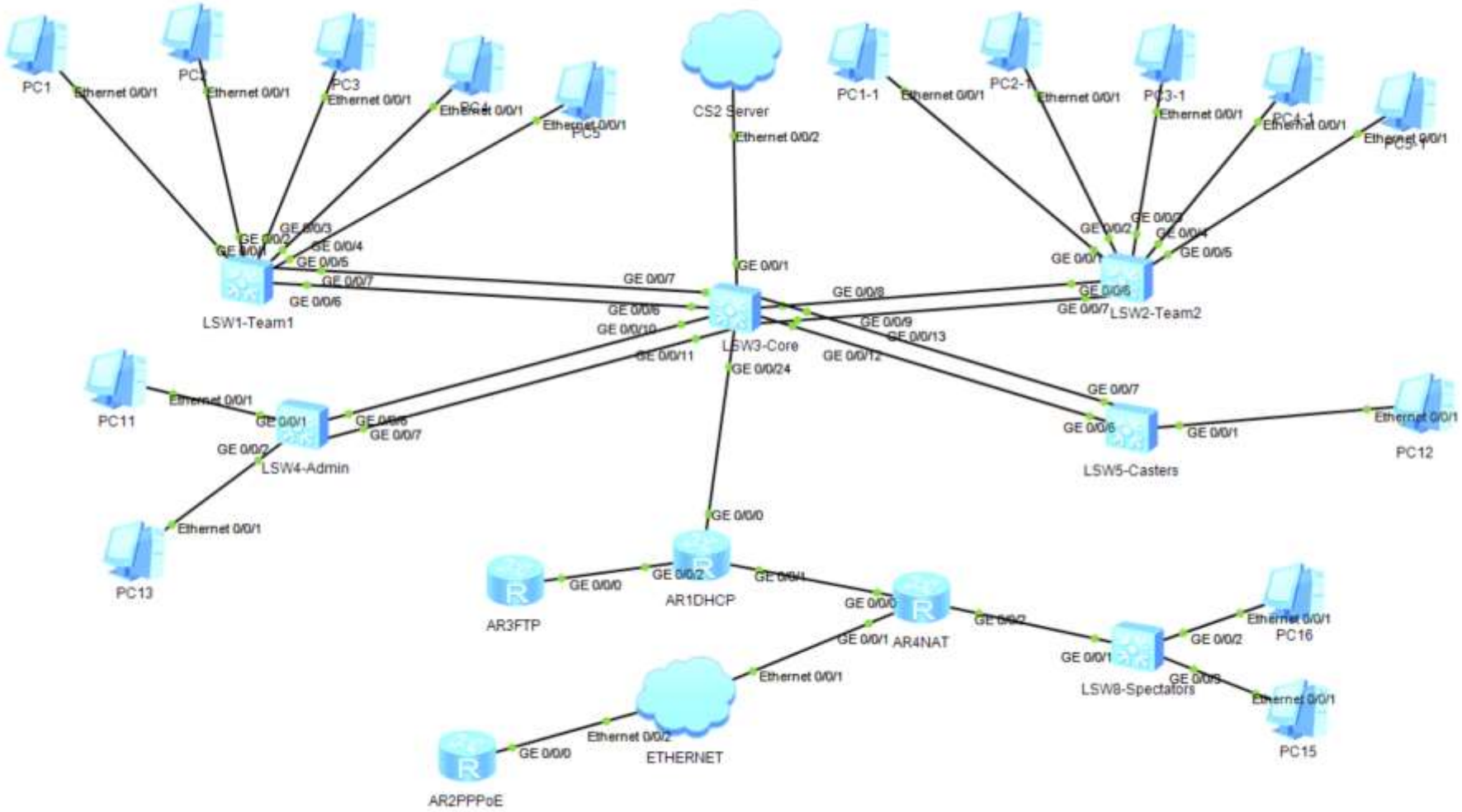
PGL Astana





This project was inspired by PGL Astana, one of the largest CS2 esports tournaments. The goal was to design a realistic tournament network infrastructure that supports players, administrators, casters, spectators, and game servers.

Topology



The network was designed using a hierarchical topology with a central Core Switch. Separate VLANs were created for players, administrators, casters, spectators, and servers to simplify management and improve security.

Component	Name / Role	IPv4	IPv6
AR1	Core Router, DHCP, OSPF	172.16.13.1 / 172.16.14.1	2001:DB8:13::1 / 2001:DB8:14::1
AR2	ISP / PPPoE Server	119.84.111.254	-
AR3	FTP Router	172.16.13.2	2001:DB8:13::2
AR4	NAT / PPPoE Client	172.16.14.2	2001:DB8:14::2
CS2 Server	Game Server	10.60.60.10	2001:DB8:60::10

VLAN	Name	IPv4 Network	Gateway	IPv6
10	Team1	10.10.10.0/24	10.10.10.1	No
20	Team2	10.20.20.0/24	10.20.20.1	No
30	Admin	10.30.30.0/24	10.30.30.1	2001:DB8:30::/64
40	Casters	10.40.40.0/24	10.40.40.1	2001:DB8:40::/64
50	Spectators	10.50.50.0/24	10.50.50.1	2001:DB8:50::/64
60	Servers	10.60.60.0/24	10.60.60.1	2001:DB8:60::/64

VLAN

```
<S3>display vlan
The total number of vlans is : 8
--------------------------------------------------------------------------------
U: Up;         D: Down;         TG: Tagged;         UT: Untagged;
MP: Vlan-mapping;               ST: Vlan-stacking;
#: ProtocolTransparent-vlan;    *: Management-vlan;
--------------------------------------------------------------------------------

VID  Type    Ports
--------------------------------------------------------------------------------
1    common  UT:GE0/0/2(D)      GE0/0/3(D)      GE0/0/4(D)      GE0/0/5(D)
             GE0/0/14(D)       GE0/0/15(D)     GE0/0/16(D)     GE0/0/17(D)
             GE0/0/18(D)       GE0/0/19(D)     GE0/0/20(D)     GE0/0/21(D)
             GE0/0/22(D)       GE0/0/23(D)     GE0/0/24(U)     Eth-Trunk1(U)
             Eth-Trunk2(U)     Eth-Trunk3(U)   Eth-Trunk4(U)

10   common  TG:GE0/0/24(U)     Eth-Trunk1(U)   Eth-Trunk2(U)   Eth-Trunk3(U)
             Eth-Trunk4(U)

20   common  TG:GE0/0/24(U)     Eth-Trunk1(U)   Eth-Trunk2(U)   Eth-Trunk3(U)
             Eth-Trunk4(U)

30   common  TG:GE0/0/24(U)     Eth-Trunk1(U)   Eth-Trunk2(U)   Eth-Trunk3(U)
             Eth-Trunk4(U)

40   common  TG:GE0/0/24(U)     Eth-Trunk1(U)   Eth-Trunk2(U)   Eth-Trunk3(U)
             Eth-Trunk4(U)

50   common  TG:GE0/0/24(U)     Eth-Trunk1(U)   Eth-Trunk2(U)   Eth-Trunk3(U)
             Eth-Trunk4(U)

60   common  UT:GE0/0/1(U)
             TG:GE0/0/24(U)     Eth-Trunk1(U)   Eth-Trunk2(U)   Eth-Trunk3(U)
             Eth-Trunk4(U)

99   common  TG:GE0/0/24(U)     Eth-Trunk1(U)   Eth-Trunk2(U)   Eth-Trunk3(U)
             Eth-Trunk4(U)

VID  Status  Property      MAC-LRN Statistics Description
--------------------------------------------------------------------------------

1    enable  default       enable  disable    VLAN 0001
10   enable  default       enable  disable    VLAN 0010
20   enable  default       enable  disable    VLAN 0020
30   enable  default       enable  disable    VLAN 0030

<S3>
```

VLAN segmentation was implemented to isolate different groups of users and services. This reduces broadcast traffic and improves overall network security.

```
<S1>display port vlan
Port                      Link Type    PVID  Trunk VLAN List
--------------------------------------------------------------------------------
Eth-Trunk1                trunk        1     1 10 20 30 40 50 60 99
GigabitEthernet0/0/1      access       10    -
GigabitEthernet0/0/2      access       10    -
GigabitEthernet0/0/3      access       10    -
GigabitEthernet0/0/4      access       10    -
GigabitEthernet0/0/5      access       10    -
GigabitEthernet0/0/6      hybrid       0     -
GigabitEthernet0/0/7      hybrid       0     -
GigabitEthernet0/0/8      hybrid       1     -
GigabitEthernet0/0/9      hybrid       1     -
GigabitEthernet0/0/10     hybrid       1     -
GigabitEthernet0/0/11     hybrid       1     -
GigabitEthernet0/0/12     hybrid       1     -
GigabitEthernet0/0/13     hybrid       1     -
GigabitEthernet0/0/14     hybrid       1     -
GigabitEthernet0/0/15     hybrid       1     -
GigabitEthernet0/0/16     hybrid       1     -
GigabitEthernet0/0/17     hybrid       1     -
GigabitEthernet0/0/18     hybrid       1     -
GigabitEthernet0/0/19     hybrid       1     -
GigabitEthernet0/0/20     hybrid       1     -
GigabitEthernet0/0/21     hybrid       1     -
GigabitEthernet0/0/22     hybrid       1     -
GigabitEthernet0/0/23     hybrid       1     -

<S1>
```

Ether-Trunk

```
<S3>display eth-trunk
Eth-Trunk1's state information is:
Local:
LAG ID: 1                   WorkingMode: STATIC
Preempt Delay: Disabled     Hash arithmetic: According to SIP-XOR-DIP
System Priority: 32768      System ID: 4c1f-ccf7-6e4a
Least Active-linknumber: 1  Max Active-linknumber: 8
Operate status: up          Number Of Up Port In Trunk: 2
--------------------------------------------------------------------------------
ActorPortName          Status   PortType PortPri PortNo PortKey PortState Weight
GigabitEthernet0/0/6   Selected 1GE      32768   7      305     10111100  1
GigabitEthernet0/0/7   Selected 1GE      32768   8      305     10111100  1

Partner:
--------------------------------------------------------------------------------
ActorPortName          SysPri   SystemID        PortPri PortNo PortKey PortState
GigabitEthernet0/0/6   32768    4c1f-ccld-470f  32768   7      305     10111100
GigabitEthernet0/0/7   32768    4c1f-ccld-470f  32768   8      305     10111100

Eth-Trunk2's state information is:
Local:
LAG ID: 2                   WorkingMode: STATIC
Preempt Delay: Disabled     Hash arithmetic: According to SIP-XOR-DIP
System Priority: 32768      System ID: 4c1f-ccf7-6e4a
Least Active-linknumber: 1  Max Active-linknumber: 8
Operate status: up          Number Of Up Port In Trunk: 2
--------------------------------------------------------------------------------
ActorPortName          Status   PortType PortPri PortNo PortKey PortState Weight
GigabitEthernet0/0/8   Selected 1GE      32768   9      561     10111100  1
GigabitEthernet0/0/9   Selected 1GE      32768   10     561     10111100  1

Partner:
--------------------------------------------------------------------------------
ActorPortName          SysPri   SystemID        PortPri PortNo PortKey PortState
GigabitEthernet0/0/8   32768    4c1f-cc79-7fba  32768   7      561     10111100
GigabitEthernet0/0/9   32768    4c1f-cc79-7fba  32768   8      561     10111100

Eth-Trunk3's state information is:
Local:
LAG ID: 3                   WorkingMode: STATIC
Preempt Delay: Disabled     Hash arithmetic: According to SIP-XOR-DIP
System Priority: 32768      System ID: 4c1f-ccf7-6e4a
Least Active-linknumber: 1  Max Active-linknumber: 8

<S3>
```

Ether-Trunk was configured between switches to increase bandwidth and provide link redundancy. If one physical connection fails, traffic continues through the remaining links.

```
<S3>display eth-trunk 1
Eth-Trunk1's state information is:
Local:
LAG ID: 1                   WorkingMode: STATIC
Preempt Delay: Disabled     Hash arithmetic: According to SIP-XOR-DIP
System Priority: 32768      System ID: 4c1f-ccf7-6e4a
Least Active-linknumber: 1  Max Active-linknumber: 8
Operate status: up          Number Of Up Port In Trunk: 2
--------------------------------------------------------------------------------
ActorPortName          Status   PortType PortPri PortNo PortKey PortState Weight
GigabitEthernet0/0/6   Selected 1GE      32768   7      305     10111100  1
GigabitEthernet0/0/7   Selected 1GE      32768   8      305     10111100  1

Partner:
--------------------------------------------------------------------------------
ActorPortName          SysPri   SystemID        PortPri PortNo PortKey PortState
GigabitEthernet0/0/6   32768    4c1f-ccld-470f  32768   7      305     10111100
GigabitEthernet0/0/7   32768    4c1f-ccld-470f  32768   8      305     10111100

<S3>
```

RSTP

RSTP was deployed to prevent Layer 2 loops and ensure stable network operation. The Core Switch was configured as the root bridge for optimal path selection.

```
<S3>
<S3>display stp
-------[CIST Global Info][Mode RSTP]-------
CIST Bridge          :0    .4c1f-ccf7-6e4a
Config Times         :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times         :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC       :0    .4c1f-ccf7-6e4a / 0
CIST RegRoot/IRPC    :0    .4c1f-ccf7-6e4a / 0
CIST RootPortId      :0.0
BPDU-Protection      :Enabled
CIST Root Type       :Primary root
TC or TCN received   :8
TC count per hello   :0
STP Converge Mode    :Normal
Time since last TC   :0 days 2h:24m:9s
Number of TC         :9
Last TC occurred     :GigabitEthernet0/0/24
----[Port5(GigabitEthernet0/0/1)][FORWARDING]----
 Port Protocol       :Enabled
 Port Role           :Designated Port
 Port Priority       :128
 Port Cost(Dot1T )   :Config=auto / Active=20000
 Designated Bridge/Port    :0.4c1f-ccf7-6e4a / 128.5
 Port Edged          :Config=enabled / Active=enabled
 BPDU-Protection     :Enabled
 Point-to-point      :Config=auto / Active=true

<S3>
```

```
<S2>display stp brief
 MSTID  Port                        Role  STP State     Protection
   0    GigabitEthernet0/0/1        DESI  FORWARDING      BPDU
   0    GigabitEthernet0/0/2        DESI  FORWARDING      BPDU
   0    GigabitEthernet0/0/3        DESI  FORWARDING      BPDU
   0    GigabitEthernet0/0/4        DESI  FORWARDING      BPDU
   0    GigabitEthernet0/0/5        DESI  FORWARDING      BPDU
   0    Eth-Trunk2                  ROOT  FORWARDING      NONE
<S2>
```

Router-on-a-Stick

Inter-VLAN routing was implemented on AR1 using Router-on-a-Stick technology. This allows devices from different VLANs to communicate through a single physical interface.

```
<AR1>display ip interface brief
*down: administratively down
^down: standby
(l): loopback
(s): spoofing
The number of interface that is UP in Physical is 11
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 10
The number of interface that is DOWN in Protocol is 1

Interface                         IP Address/Mask      Physical   Protocol
GigabitEthernet0/0/0              10.99.99.1/24        up         up
GigabitEthernet0/0/0.10           10.10.10.1/24        up         up
GigabitEthernet0/0/0.20           10.20.20.1/24        up         up
GigabitEthernet0/0/0.30           10.30.30.1/24        up         up
GigabitEthernet0/0/0.40           10.40.40.1/24        up         up
GigabitEthernet0/0/0.50           10.50.50.1/24        up         up
GigabitEthernet0/0/0.60           10.60.60.1/24        up         up
GigabitEthernet0/0/0.99           unassigned           up         down
GigabitEthernet0/0/1              172.16.14.1/30       up         up
GigabitEthernet0/0/2              172.16.13.1/30       up         up
NULL0                             unassigned           up         up(s)
<AR1>
```

```
<AR1>display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 32        Routes : 32

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

        0.0.0.0/0   O_ASE   150  1           D   172.16.14.2     GigabitEthernet
0/0/1
     10.10.10.0/24  Direct  0    0           D   10.10.10.1      GigabitEthernet
0/0/0.10
     10.10.10.1/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0.10
   10.10.10.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0.10
     10.20.20.0/24  Direct  0    0           D   10.20.20.1      GigabitEthernet
0/0/0.20
     10.20.20.1/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0.20
   10.20.20.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0.20
     10.30.30.0/24  Direct  0    0           D   10.30.30.1      GigabitEthernet
0/0/0.30
     10.30.30.1/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0.30
   10.30.30.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0.30
     10.40.40.0/24  Direct  0    0           D   10.40.40.1      GigabitEthernet
0/0/0.40
     10.40.40.1/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0.40
   10.40.40.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0.40
     10.50.50.0/24  Direct  0    0           D   10.50.50.1      GigabitEthernet
0/0/0.50
     10.50.50.1/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0.50
   10.50.50.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0.50
     10.60.60.0/24  Direct  0    0           D   10.60.60.1      GigabitEthernet
0/0/0.60
     10.60.60.1/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0.60
   10.60.60.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0.60
     10.99.99.0/24  Direct  0    0           D   10.99.99.1      GigabitEthernet
0/0/0
     10.99.99.1/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
   10.99.99.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
127.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0
    172.16.13.0/30  Direct  0    0           D   172.16.13.1     GigabitEthernet
0/0/2
    172.16.13.1/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
```

DHCP

```
<AR1>display ip pool
  ------------------------------------------------------------------------
  Pool-name      : VLAN10
  Pool-No        : 0
  Position       : Local          Status          : Unlocked
  Gateway-0      : 10.10.10.1
  Mask           : 255.255.255.0
  VPN instance   : --

  ------------------------------------------------------------------------
  Pool-name      : VLAN20
  Pool-No        : 1
  Position       : Local          Status          : Unlocked
  Gateway-0      : 10.20.20.1
  Mask           : 255.255.255.0
  VPN instance   : --

  ------------------------------------------------------------------------
  Pool-name      : VLAN30
  Pool-No        : 2
  Position       : Local          Status          : Unlocked
  Gateway-0      : 10.30.30.1
  Mask           : 255.255.255.0
  VPN instance   : --

  ------------------------------------------------------------------------
  Pool-name      : VLAN40
  Pool-No        : 3
  Position       : Local          Status          : Unlocked
  Gateway-0      : 10.40.40.1
  Mask           : 255.255.255.0
  VPN instance   : --

  ------------------------------------------------------------------------
  Pool-name      : VLAN50
  Pool-No        : 4
  Position       : Local          Status          : Unlocked
  Gateway-0      : 10.50.50.1
  Mask           : 255.255.255.0
  VPN instance   : --


  IP address Statistic
    Total       :1265
    Used        :4          Idle        :1256
    Expired     :0          Conflict    :0          Disable   :5
<AR1>
```

PC3

Basic Config | Command | MCPacket | UdpPacket | Console

```
Welcome to use PC Simulator!

PC>

PC Simulator has not been started!

Welcome to use PC Simulator!

PC>ipconfig

Link local IPv6 address.....: fe80::5689:98ff:fedd:310a
IPv6 address................: :: / 128
IPv6 gateway................: ::
IPv4 address................: 10.10.10.13
Subnet mask.................: 255.255.255.0
Gateway.....................: 10.10.10.1
Physical address............: 54-89-98-DD-31-0A
DNS server..................:

PC>
```

DHCP was configured to automatically provide IP addresses and network parameters to clients. This simplifies deployment and reduces manual configuration.

OSPF

OSPF was implemented to dynamically exchange routes between routers. This removes the need for static routing and allows the network to adapt automatically.

```
<AR1>display ospf peer

      OSPF Process 1 with Router ID 1.1.1.1
            Neighbors

 Area 0.0.0.0 interface 172.16.14.1(GigabitEthernet0/0/1)'s neighbors
 Router ID: 4.4.4.4          Address: 172.16.14.2
   State: Full  Mode:Nbr is  Master  Priority: 1
   DR: 172.16.14.2  BDR: 172.16.14.1  MTU: 0
   Dead timer due in 36  sec
   Retrans timer interval: 5
   Neighbor is up for 04:11:24
   Authentication Sequence: [ 0 ]

            Neighbors

 Area 0.0.0.0 interface 172.16.13.1(GigabitEthernet0/0/2)'s neighbors
 Router ID: 3.3.3.3          Address: 172.16.13.2
   State: Full  Mode:Nbr is  Master  Priority: 1
   DR: 172.16.13.2  BDR: 172.16.13.1  MTU: 0
   Dead timer due in 35  sec
   Retrans timer interval: 5
   Neighbor is up for 04:11:25
   Authentication Sequence: [ 0 ]

<AR1>
```

```
<AR3>display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 15        Routes : 15

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

        0.0.0.0/0   O_ASE   150  1           D   172.16.13.1     GigabitEthernet
0/0/0
     10.10.10.0/24  OSPF    10   2           D   172.16.13.1     GigabitEthernet
0/0/0
     10.20.20.0/24  OSPF    10   2           D   172.16.13.1     GigabitEthernet
0/0/0
     10.30.30.0/24  OSPF    10   2           D   172.16.13.1     GigabitEthernet
0/0/0
     10.40.40.0/24  OSPF    10   2           D   172.16.13.1     GigabitEthernet
0/0/0
     10.50.50.0/24  OSPF    10   2           D   172.16.13.1     GigabitEthernet
0/0/0
     10.60.60.0/24  OSPF    10   2           D   172.16.13.1     GigabitEthernet
0/0/0
      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
127.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0
    172.16.13.0/30  Direct  0    0           D   172.16.13.2     GigabitEthernet
0/0/0
    172.16.13.2/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
    172.16.13.3/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
    172.16.14.0/30  OSPF    10   2           D   172.16.13.1     GigabitEthernet
0/0/0
255.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0

<AR3>
```

ACL

Access Control Lists were configured to enforce security policies. User access to network resources is controlled according to their role within the tournament infrastructure.

```
<AR1>display acl all
 Total quantity of nonempty ACL number is 5

Advanced ACL 3000, 6 rules
Acl's step is 5
 rule 5 deny ip source 10.50.50.0 0.0.0.255 destination 10.10.10.0 0.0.0.255
 rule 10 deny ip source 10.50.50.0 0.0.0.255 destination 10.20.20.0 0.0.0.255
 rule 15 deny ip source 10.50.50.0 0.0.0.255 destination 10.30.30.0 0.0.0.255
 rule 20 deny ip source 10.50.50.0 0.0.0.255 destination 10.40.40.0 0.0.0.255
 rule 25 deny ip source 10.50.50.0 0.0.0.255 destination 10.60.60.0 0.0.0.255
 rule 30 permit ip

Advanced ACL 3001, 2 rules
Acl's step is 5
 rule 5 deny ip source 10.10.10.0 0.0.0.255 destination 203.0.113.10 0
 rule 10 permit ip (3 matches)

Advanced ACL 3002, 2 rules
Acl's step is 5
 rule 5 deny ip source 10.20.20.0 0.0.0.255 destination 203.0.113.10 0
 rule 10 permit ip

Advanced ACL 3003, 6 rules
Acl's step is 5
 rule 5 permit ip source 10.10.10.0 0.0.0.255 destination 10.60.60.10 0
 rule 10 permit ip source 10.20.20.0 0.0.0.255 destination 10.60.60.10 0
 rule 15 permit ip source 10.30.30.0 0.0.0.255 destination 10.60.60.0 0.0.0.255
 rule 20 permit ip source 10.40.40.0 0.0.0.255 destination 10.60.60.10 0
 rule 25 deny ip destination 10.60.60.0 0.0.0.255
 rule 30 permit ip

Advanced ACL 3004, 3 rules
Acl's step is 5
 rule 5 deny ip source 10.40.40.0 0.0.0.255 destination 10.30.30.0 0.0.0.255
 rule 10 permit ip source 10.40.40.0 0.0.0.255 destination 10.60.60.10 0
 rule 15 permit ip (3 matches)

<AR1>
```

PC16

Basic Config | Command | MCPacket | UdpPacket | Console

```
Welcome to use PC Simulator!

PC>

PC Simulator has not been started!

Welcome to use PC Simulator!

PC>ping 10.60.60.10

Ping 10.60.60.10: 32 data bytes, Press Ctrl_C to break
From 0.0.0.0: Destination host unreachable
From 0.0.0.0: Destination host unreachable
From 0.0.0.0: Destination host unreachable
From 0.0.0.0: Destination host unreachable
From 0.0.0.0: Destination host unreachable

--- 10.60.60.10 ping statistics ---
  5 packet(s) transmitted
  0 packet(s) received
  100.00% packet loss

PC>
```

PC11

Basic Config | Command | MCPacket | UdpPacket | Console

```
Physical address.....: 54-89-98-43-4C-70
DNS server.....: 8.8.8.8
                 1.1.1.1

PC>

PC Simulator has not been started!

Welcome to use PC Simulator!

PC>ping 10.60.60.10

Ping 10.60.60.10: 32 data bytes, Press Ctrl_C to break
From 10.60.60.10: bytes=32 seq=1 ttl=127 time=109 ms
From 10.60.60.10: bytes=32 seq=2 ttl=127 time=94 ms
From 10.60.60.10: bytes=32 seq=3 ttl=127 time=93 ms
From 10.60.60.10: bytes=32 seq=4 ttl=127 time=110 ms
From 10.60.60.10: bytes=32 seq=5 ttl=127 time=78 ms

--- 10.60.60.10 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 78/96/110 ms

PC>
```

FTP

An FTP server was deployed on AR3 to store configuration backups and project files. Centralized storage simplifies administration and recovery procedures.

```
<AR1>ftp 172.16.13.2
Trying 172.16.13.2 ...

Press CTRL+K to abort
Connected to 172.16.13.2.
220 FTP service ready.
User(172.16.13.2:(none)):ftpadmin
331 Password required for ftpadmin.
Enter password:
230 User logged in.

[AR1-ftp]dir
200 Port command okay.
150 Opening ASCII mode data connection for *.
drwxrwxrwx    1 noone    nogroup          0 May 30 13:21 dhcp
-rwxrwxrwx    1 noone    nogroup     121802 May 26  2014 portalpage.zip
-rwxrwxrwx    1 noone    nogroup       2263 May 30 15:26 statemach.efs
-rwxrwxrwx    1 noone    nogroup     828482 May 26  2014 sslvpn.zip
-rwxrwxrwx    1 noone    nogroup        267 May 30 15:25 private-data.txt
drwxrwxrwx    1 noone    nogroup          0 May 30 15:26 .
-rwxrwxrwx    1 noone    nogroup        877 May 30 15:26 vrpcfg.zip
226 Transfer complete.
FTP: 467 byte(s) received in 0.040 second(s) 11.67Kbyte(s)/sec.

[AR1-ftp]
```

PPPoE

```
<AR4>display interface Dialer1
Dialer1 current state : UP
Line protocol current state : UP (spoofing)
Description:HUAWEI, AR Series, Dialer1 Interface
Route Port,The Maximum Transmit Unit is 1500, Hold timer is 10(sec)
Internet Address is negotiated, 119.84.111.253/32
Link layer protocol is PPP
LCP initial
Physical is Dialer
Current system time: 2026-05-31 03:47:24-08:00
    Last 300 seconds input rate 0 bits/sec, 0 packets/sec
    Last 300 seconds output rate 0 bits/sec, 0 packets/sec
    Realtime 0 seconds input rate 0 bits/sec, 0 packets/sec
    Realtime 0 seconds output rate 0 bits/sec, 0 packets/sec
    Input: 0 bytes
    Output:0 bytes
    Input bandwidth utilization  :    0%
    Output bandwidth utilization :    0%
Bound to Dialer1:0:
Dialer1:0 current state : UP ,
Line protocol current state : UP

Link layer protocol is PPP
LCP opened, IPCP opened
Packets statistics:
  Input packets:0,   0 bytes
  Output packets:0, 0 bytes
  FCS error packets:0
  Address error packets:0
  Control field control error packets:0


<AR4>
```

PPPoE was configured to simulate an Internet Service Provider connection. AR4 establishes a client session and receives network parameters dynamically.

```
<AR4>display ip interface brief
*down: administratively down
^down: standby
(l): loopback
(s): spoofing
The number of interface that is UP in Physical is 6
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 4
The number of interface that is DOWN in Protocol is 2

Interface                         IP Address/Mask      Physical   Protocol
Dialer1                           119.84.111.253/32    up         up(s)
GigabitEthernet0/0/0              172.16.14.2/30       up         up
GigabitEthernet0/0/1              unassigned           up         down
GigabitEthernet0/0/2              unassigned           up         down
LoopBack0                         unassigned           up         up(s)
NULL0                             unassigned           up         up(s)
<AR4>
```

NAT

NAT was configured on AR4 to translate private VLAN addresses into a public PPPoE address, enabling internal networks to access external resources through a single WAN connection.

```
<AR4>display current-configuration interface Dialer1
[V200R003C00]
#
interface Dialer1
 link-protocol ppp
 ppp chap user ar4
 ppp chap password simple Huawei123
 ip address ppp-negotiate
 dialer user ar4
 dialer bundle 1
 dialer-group 1
 nat outbound 2000
#
return
<AR4>
```

```
<AR4>display nat session all
  NAT Session Table Information:

  Total : 0
<AR4>
```

```
<AR4>display acl 2000
Basic ACL 2000, 6 rules
Acl's step is 5
 rule 5 permit source 10.10.10.0 0.0.0.255
 rule 10 permit source 10.20.20.0 0.0.0.255
 rule 15 permit source 10.30.30.0 0.0.0.255
 rule 20 permit source 10.40.40.0 0.0.0.255
 rule 25 permit source 10.50.50.0 0.0.0.255
 rule 30 permit source 10.60.60.0 0.0.0.255

<AR4>
```

IPv6

```
<AR1>display ipv6 interface brief
*down: administratively down
(l): loopback
(s): spoofing
Interface                   Physical              Protocol
GigabitEthernet0/0/0.30     up                    up
[IPv6 Address] 2001:DB8:30::1
GigabitEthernet0/0/0.40     up                    up
[IPv6 Address] 2001:DB8:40::1
GigabitEthernet0/0/0.50     up                    up
[IPv6 Address] 2001:DB8:50::1
GigabitEthernet0/0/0.60     up                    up
[IPv6 Address] 2001:DB8:60::1
GigabitEthernet0/0/1        up                    up
[IPv6 Address] 2001:DB8:14::1
GigabitEthernet0/0/2        up                    up
[IPv6 Address] 2001:DB8:13::1
<AR1>
```

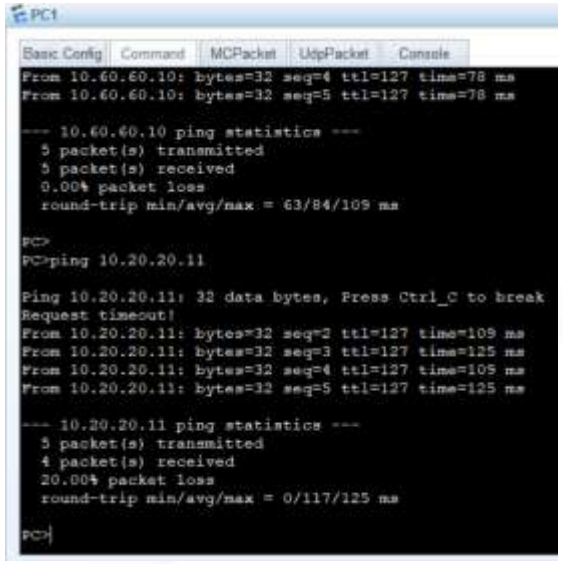
IPv6 addressing and OSPFv3 routing were deployed alongside IPv4. This demonstrates support for modern networking standards and dual-stack operation.

```
<AR1>display ospfv3 peer
OSPFv3 Process (10)
OSPFv3 Area (0.0.0.0)
Neighbor ID     Pri  State           Dead Time Interface          Instance ID
4.4.4.4           1  Full/DR         00:00:36  GE0/0/1                    0
3.3.3.3           1  Full/DR         00:00:33  GE0/0/2                    0

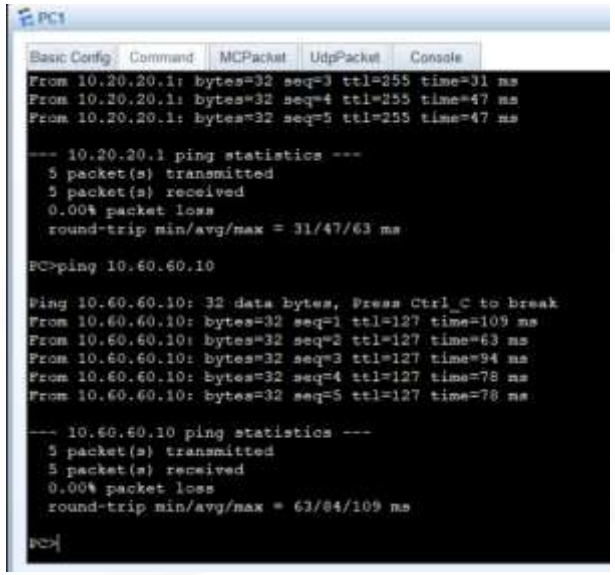
<AR1>
```

PING

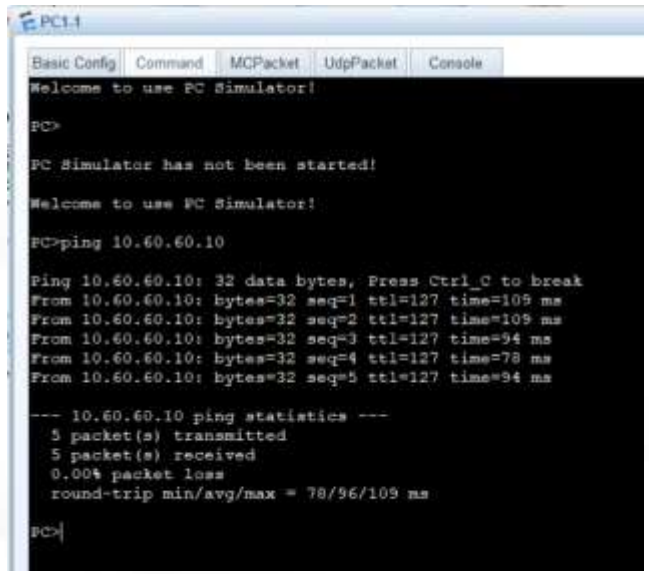
Team1 → Team2



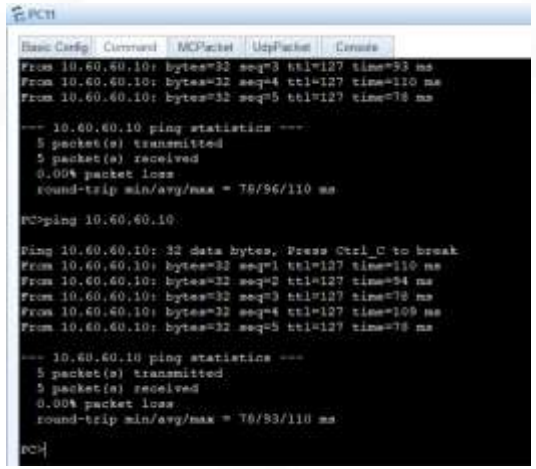
Team1 → CS2 Server



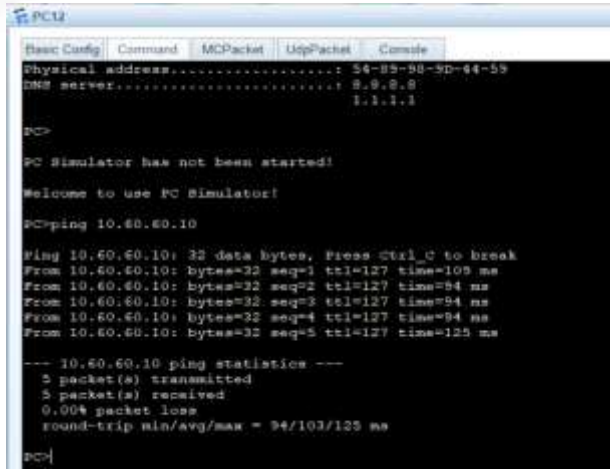
Team2 → CS2 Server



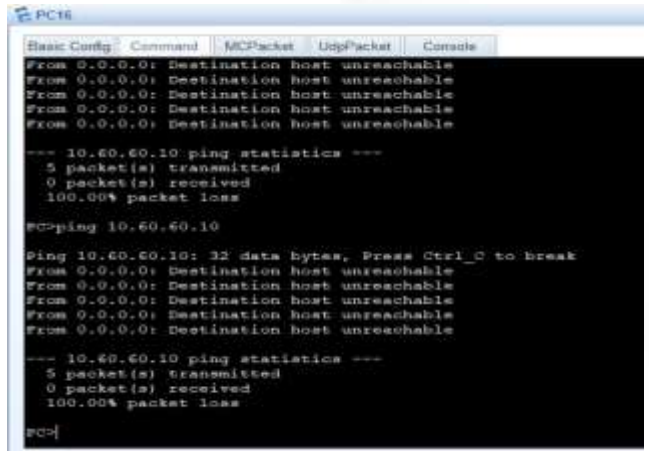
Admin → CS2 Server



Casters → CS2 Server

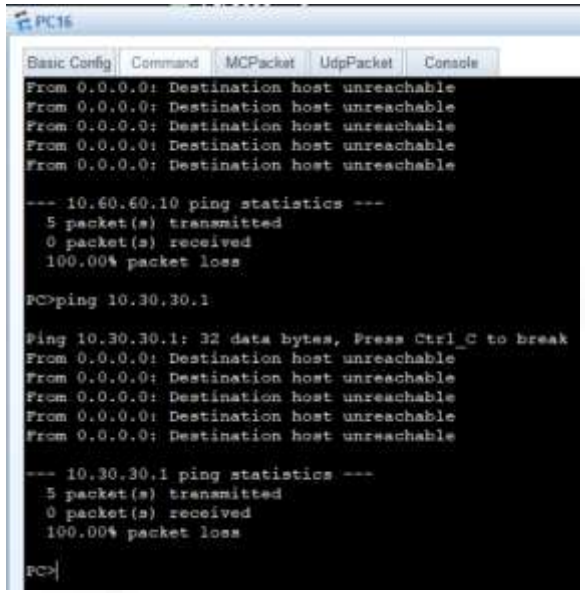


Spectators → CS2 Server

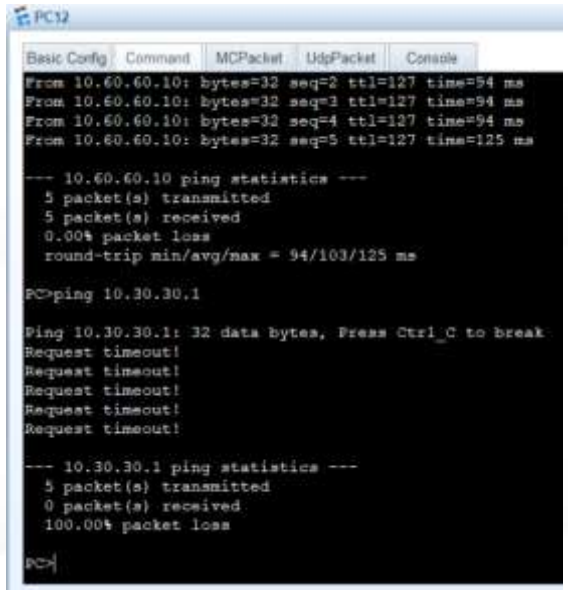


PING (ACL,OSPF, PPPoE,IPv6)

Spectators → Admin



Casters → Admin



AR3 → VLAN60 (OSPF)

```
<AR3>
<AR3>ping 10.60.60.10
  PING 10.60.60.10: 56  data bytes, press CTRL_C to break
    Reply from 10.60.60.10: bytes=56 Sequence=1 ttl=127 time=80 ms
    Reply from 10.60.60.10: bytes=56 Sequence=2 ttl=127 time=30 ms
    Reply from 10.60.60.10: bytes=56 Sequence=3 ttl=127 time=50 ms
    Reply from 10.60.60.10: bytes=56 Sequence=4 ttl=127 time=40 ms
    Reply from 10.60.60.10: bytes=56 Sequence=5 ttl=127 time=30 ms

  --- 10.60.60.10 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 30/46/80 ms

<AR3>
```

OSPF distributes routes.

AR4 → Admin VLAN

```
<AR4>ping 10.30.30.1
  PING 10.30.30.1: 56  data bytes, press CTRL_C to break
    Reply from 10.30.30.1: bytes=56 Sequence=1 ttl=255 time=50 ms
    Reply from 10.30.30.1: bytes=56 Sequence=2 ttl=255 time=20 ms
    Reply from 10.30.30.1: bytes=56 Sequence=3 ttl=255 time=30 ms
    Reply from 10.30.30.1: bytes=56 Sequence=4 ttl=255 time=20 ms
    Reply from 10.30.30.1: bytes=56 Sequence=5 ttl=255 time=30 ms

  --- 10.30.30.1 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 20/30/50 ms

<AR4>
```

The OSPF between AR1 and AR4 works.

AR4 → AR2(PPPoE)

```
<AR4>ping 119.84.111.254
  PING 119.84.111.254: 56  data bytes, press CTRL_C to break
    Reply from 119.84.111.254: bytes=56 Sequence=1 ttl=255 time=90 ms
    Reply from 119.84.111.254: bytes=56 Sequence=2 ttl=255 time=20 ms
    Reply from 119.84.111.254: bytes=56 Sequence=3 ttl=255 time=40 ms
    Reply from 119.84.111.254: bytes=56 Sequence=4 ttl=255 time=30 ms
    Reply from 119.84.111.254: bytes=56 Sequence=5 ttl=255 time=20 ms

  --- 119.84.111.254 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 20/40/90 ms

<AR4>
```

AR1 → AR3(IPv6)

```
<AR1>ping ipv6 2001:DB8:13::2
  PING 2001:DB8:13::2 : 56  data bytes, press CTRL_C to break
    Reply from 2001:DB8:13::2
    bytes=56 Sequence=1 hop limit=64  time = 50 ms
    Reply from 2001:DB8:13::2
    bytes=56 Sequence=2 hop limit=64  time = 30 ms
    Reply from 2001:DB8:13::2
    bytes=56 Sequence=3 hop limit=64  time = 20 ms
    Reply from 2001:DB8:13::2
    bytes=56 Sequence=4 hop limit=64  time = 20 ms
    Reply from 2001:DB8:13::2
    bytes=56 Sequence=5 hop limit=64  time = 20 ms

  --- 2001:DB8:13::2 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 20/28/50 ms

<AR1>
```

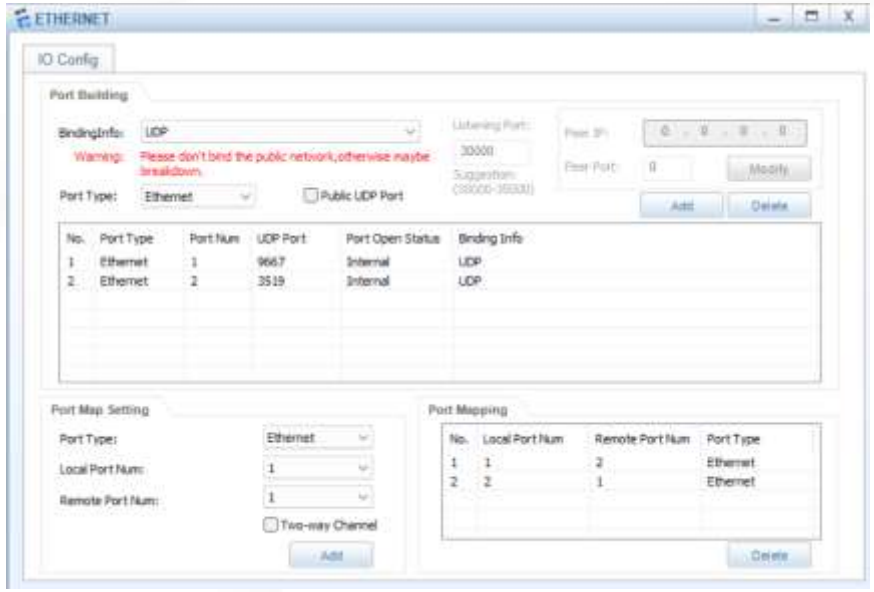
PING

AR1 → AR4(IPv6)

```
<AR1>ping ipv6 2001:DB8:14::2
  PING 2001:DB8:14::2 : 56  data bytes, press CTRL_C to break
    Reply from 2001:DB8:14::2
    bytes=56 Sequence=1 hop limit=64  time = 60 ms
    Reply from 2001:DB8:14::2
    bytes=56 Sequence=2 hop limit=64  time = 10 ms
    Reply from 2001:DB8:14::2
    bytes=56 Sequence=3 hop limit=64  time = 30 ms
    Reply from 2001:DB8:14::2
    bytes=56 Sequence=4 hop limit=64  time = 30 ms
    Reply from 2001:DB8:14::2
    bytes=56 Sequence=5 hop limit=64  time = 20 ms

  --- 2001:DB8:14::2 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 10/30/60 ms

<AR1>
```



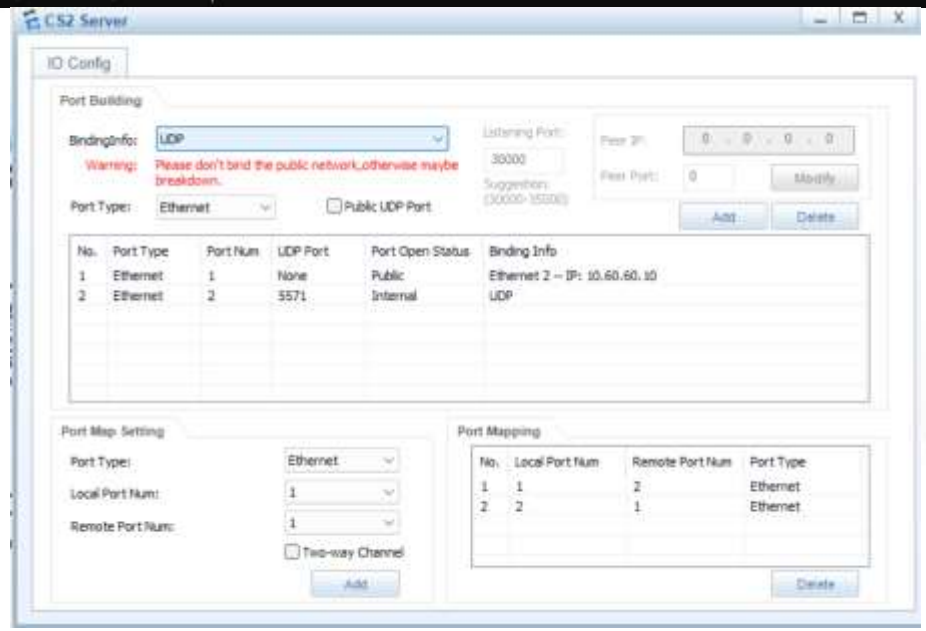
CS2 Server → Default Gateway (AR1)

```
C:\Users\kamal>ping 10.60.60.1

Обмен пакетами с 10.60.60.1 по с 32 байтами данных:
Ответ от 10.60.60.1: число байт=32 время=30мс TTL=255
Ответ от 10.60.60.1: число байт=32 время=38мс TTL=255
Ответ от 10.60.60.1: число байт=32 время=26мс TTL=255
Ответ от 10.60.60.1: число байт=32 время=34мс TTL=255

Статистика Ping для 10.60.60.1:
    Пакетов: отправлено = 4, получено = 4, потеряно = 0
    (0% потерь)
Приблизительное время приема-передачи в мс:
    Минимальное = 26мсек, Максимальное = 38 мсек, Среднее = 32 мсек

C:\Users\kamal>
```




THANK
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